

FIRE AUDIT ANALYSIS DASHBOARD

Business Case & Project Documentation

EXECUTIVE SUMMARY

The Fire Audit Analysis Dashboard is an end-to-end data analytics solution designed to provide real-time visibility into supplier fire safety audit execution. Built using Python, PostgreSQL, and Power BI, the dashboard transforms static audit records into actionable insights that help safety officers, procurement teams, and senior management monitor compliance, identify risk concentrations, and prioritize follow-up actions across the supply chain.

BUSINESS PROBLEM

Organizations conduct fire safety audits on their suppliers on a planned schedule. However, planning an audit and executing it on time are two very different operational realities. The current process suffers from several gaps:

1. No central visibility into audit execution status across the supply chain.
2. Delays between planned and actual audit dates often extend up to five months, leaving suppliers unmonitored for prolonged periods.
3. High-risk suppliers cannot be prioritized for follow-up due to the absence of consolidated risk reporting.
4. Year-on-year audit performance comparisons are unavailable, making it difficult to assess whether execution capability is scaling with audit volume.
5. Pending audits and overdue items accumulate silently with no automated alerting or visualization.

The result is operational blind spots that expose the organization to undetected fire safety risks across its supplier ecosystem.

PROJECT OBJECTIVES

- Build a unified data pipeline that ingests raw audit data and transforms it into a structured, query-ready format.
- Develop analytical SQL views that pre-compute the core business metrics required for dashboard reporting.
- Deliver a Power BI dashboard with interactive filters, KPI cards, and visual analytics that align with executive reporting standards.
- Enable data-driven prioritization of supplier follow-ups based on risk category, audit delay, and completion status.
- Establish a scalable architecture that can accommodate new audit records and expanded reporting requirements over time.

SOLUTION ARCHITECTURE

The solution follows a four-stage data pipeline:

Stage 1 — Data Cleaning (Jupyter Notebook + Python)

Raw Excel audit data is cleaned, validated, and enriched. Date columns are standardized, risk categories are normalized, derived fields such as audit status, days taken, and financial year are engineered.

Stage 2 — Data Storage (PostgreSQL)

The cleaned dataset is loaded into a PostgreSQL database using SQLAlchemy, providing a single source of truth that downstream tools connect to.

Stage 3 — Analytical Views (SQL)

Five PostgreSQL views are created to pre-aggregate key business

metrics: monthly audit performance, cumulative execution gaps, risk distribution, supplier details, and KPI summary statistics.

Stage 4 — Visualization (Power BI)

Power BI connects directly to the PostgreSQL views and renders an interactive dashboard with slicers, KPI cards, charts, and detail tables that update dynamically based on user selections.

KEY INSIGHTS DELIVERED

The dashboard surfaces several actionable findings from the audit dataset:

- Out of 45 planned audits, 35 have been completed and 10 remain pending — a completion rate of 77.8 percent.
- 51 percent of completed audits fall under the Caution Risk category and 17 percent under High Risk, indicating that the majority of audited suppliers are not fully safe.
- The longest audit delay recorded is 152 days, demonstrating significant gaps between planned and actual audit dates.
- High Risk suppliers consistently score between 52 and 73 percent, while Safe suppliers score 89 percent and above — suggesting a clear scoring threshold around the 75 percent mark.
- FY24-25 achieved a 100 percent completion rate, while FY25-26 is trailing at approximately 70 percent, signaling that audit volume has grown faster than execution capacity.

DASHBOARD CAPABILITIES

The final Power BI dashboard includes the following components:

- Three interactive slicers for Financial Year, Supplier Name, and Risk Category.

- Four KPI cards displaying Total Planned, Total Audited, Total Pending, and Completion Rate percentage.
- A Monthly Wise Plan chart comparing planned audits against actual completions and pending items.
- A Cumulative Plan chart visualizing cumulative planned audits against year-to-date audited completions.
- A Risk Category donut chart breaking down supplier distribution by Safe, Caution Risk, and High Risk categories.
- A supplier-level detail table showing audit dates, scores, days taken, and risk classification, fully filterable through the slicers above.

TECHNICAL STACK

- Python 3.13 — Data cleaning and exploratory analysis
- Pandas, NumPy — Data manipulation
- Matplotlib, Seaborn — Statistical visualization
- PostgreSQL 17 — Relational database management
- SQLAlchemy with psycopg2 — Python-to-database connectivity
- Power BI Desktop — Dashboard and visualization
- Jupyter Notebook — Analysis environment
- Git and GitHub — Version control and project hosting

BUSINESS VALUE

This project delivers measurable value to the organization in four key areas:

1. Operational Efficiency

Centralized audit tracking eliminates manual reporting and reduces the time required to assess supplier compliance.

2. Risk Mitigation

Real-time visibility into pending audits and high-risk

suppliers enables proactive intervention before incidents occur.

3. Strategic Decision-Making

Year-on-year performance benchmarking helps leadership identify capacity bottlenecks and resource allocation needs.

4. Audit Accountability

Transparent execution metrics hold internal teams and external partners accountable to agreed audit schedules.

PROJECT DELIVERABLES

The complete project is hosted on GitHub and includes:

- Jupyter Notebook with full data cleaning and EDA workflow
- Cleaned dataset in CSV format
- PostgreSQL view definitions in SQL
- Power BI dashboard file (.pbix)
- Exploratory analysis charts in PNG format
- README documentation with setup instructions

GitHub Repository:

<https://github.com/yashrajouria-lgtm/Fire-Audit-Analysis-Dashboard>

PROJECT AUTHOR

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